EXPERIMENT-12:

AIM:

To write a C program to encrypt and decrypt a message using the Hill cipher with the key matrix.

CODE:

#include <stdio.h> #include <ctype.h> #include <string.h>

int main() { char p[]="meet me at the usual place at ten rather than eight oclock"; char c[200]; int i, a, b, x, y;

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for(i=0; p[i]; i++)
    if(isalpha(p[i])) p[i]=tolower(p[i]);

printf("Plaintext: %s\n", p);
printf("Ciphertext: ");

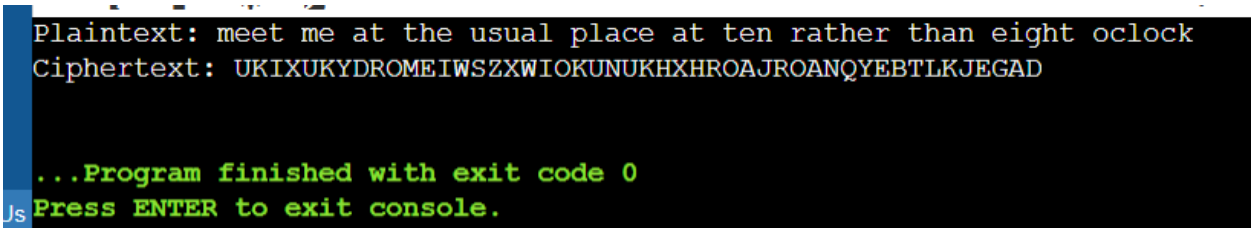
for(i=0; p[i]; ) {
    if(!isalpha(p[i])) { i++; continue; }
    a=p[i++]-'a';
    while(p[i] && !isalpha(p[i])) i++;
    if(!p[i]) { b=('x'-'a'); }
    else b=p[i++]-'a';

    x=(9*a+4*b)%26;
    y=(5*a+7*b)%26;

    printf("%c%c",'A'+x,'A'+y);
}
printf("\n");
return 0;


}
```

OUTPUT:


Plaintext: meet me at the usual place at ten rather than eight oclock
Ciphertext: UKIXUKYDROMEIWSZXWIOKUNUKHXHROAJROANQYEBTLKJEGAD
...Program finished with exit code 0
Press ENTER to exit console.